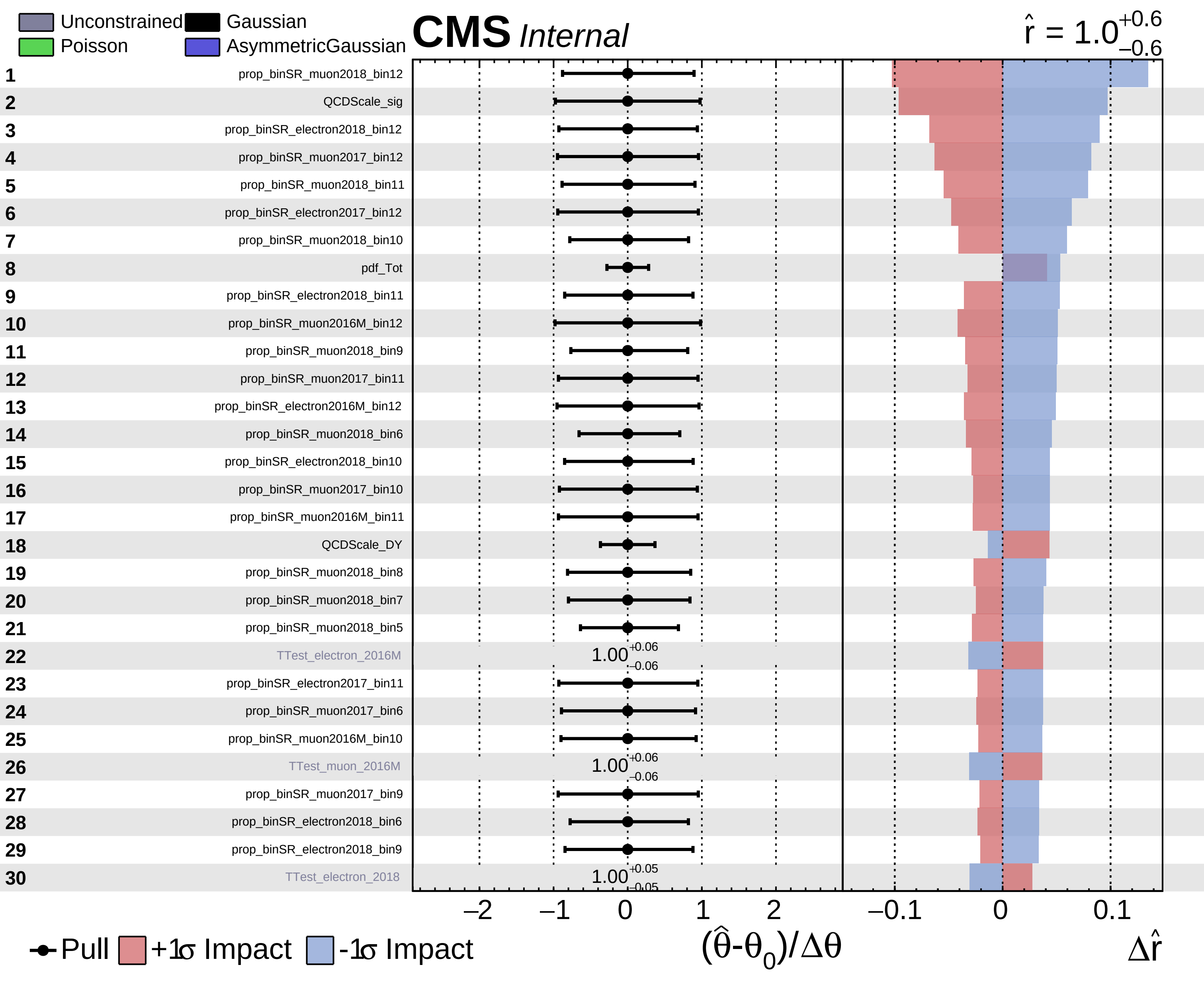
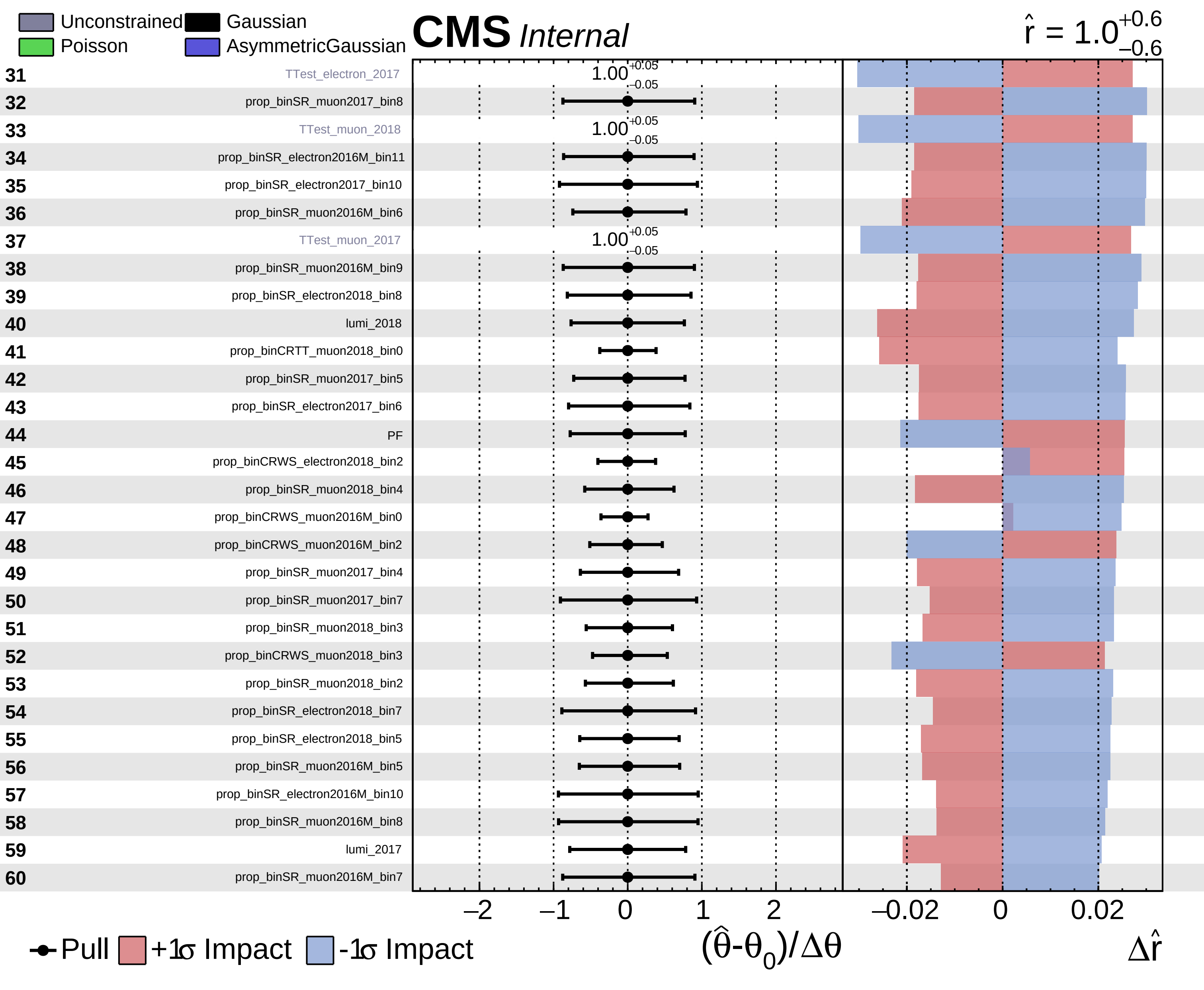
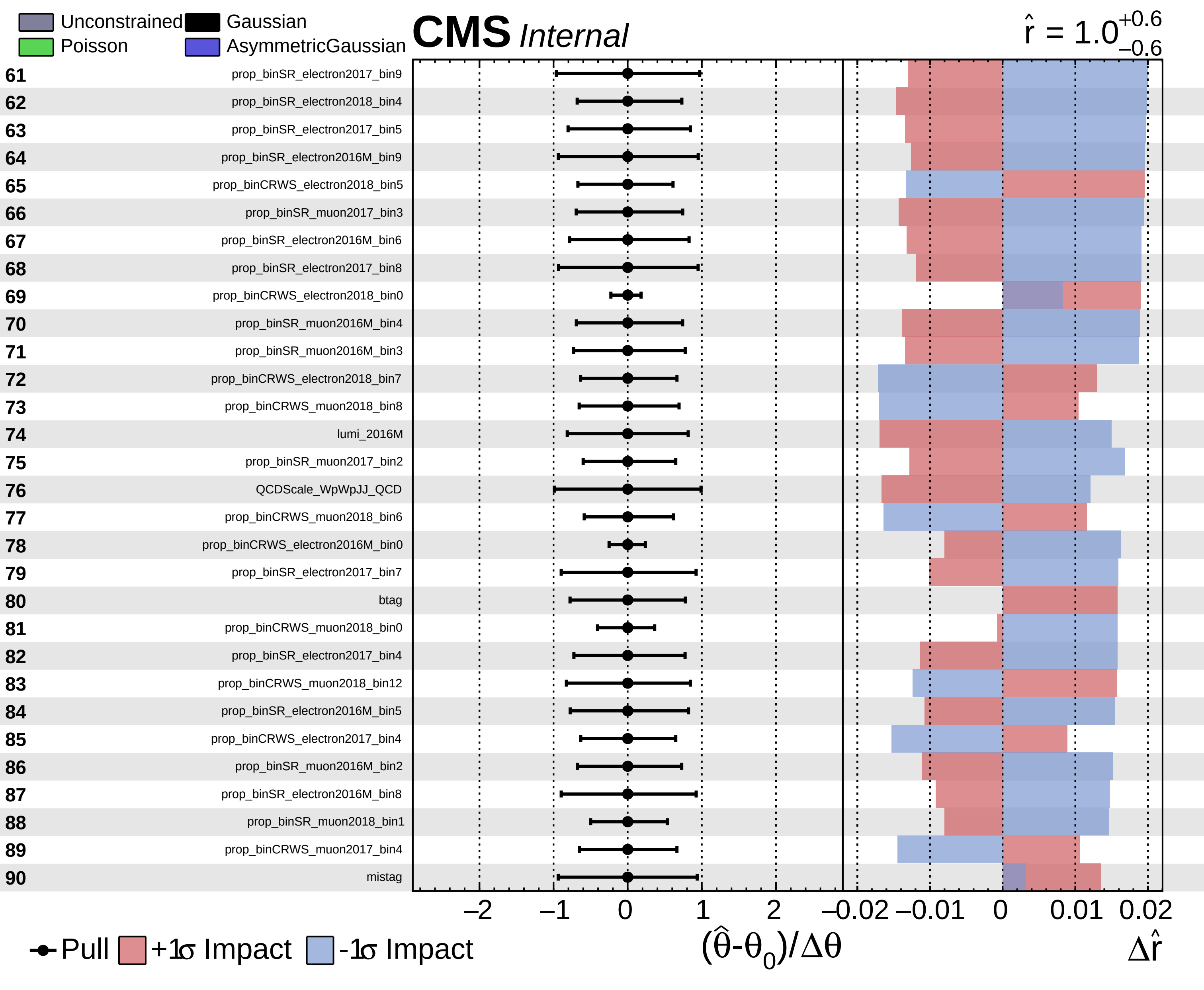
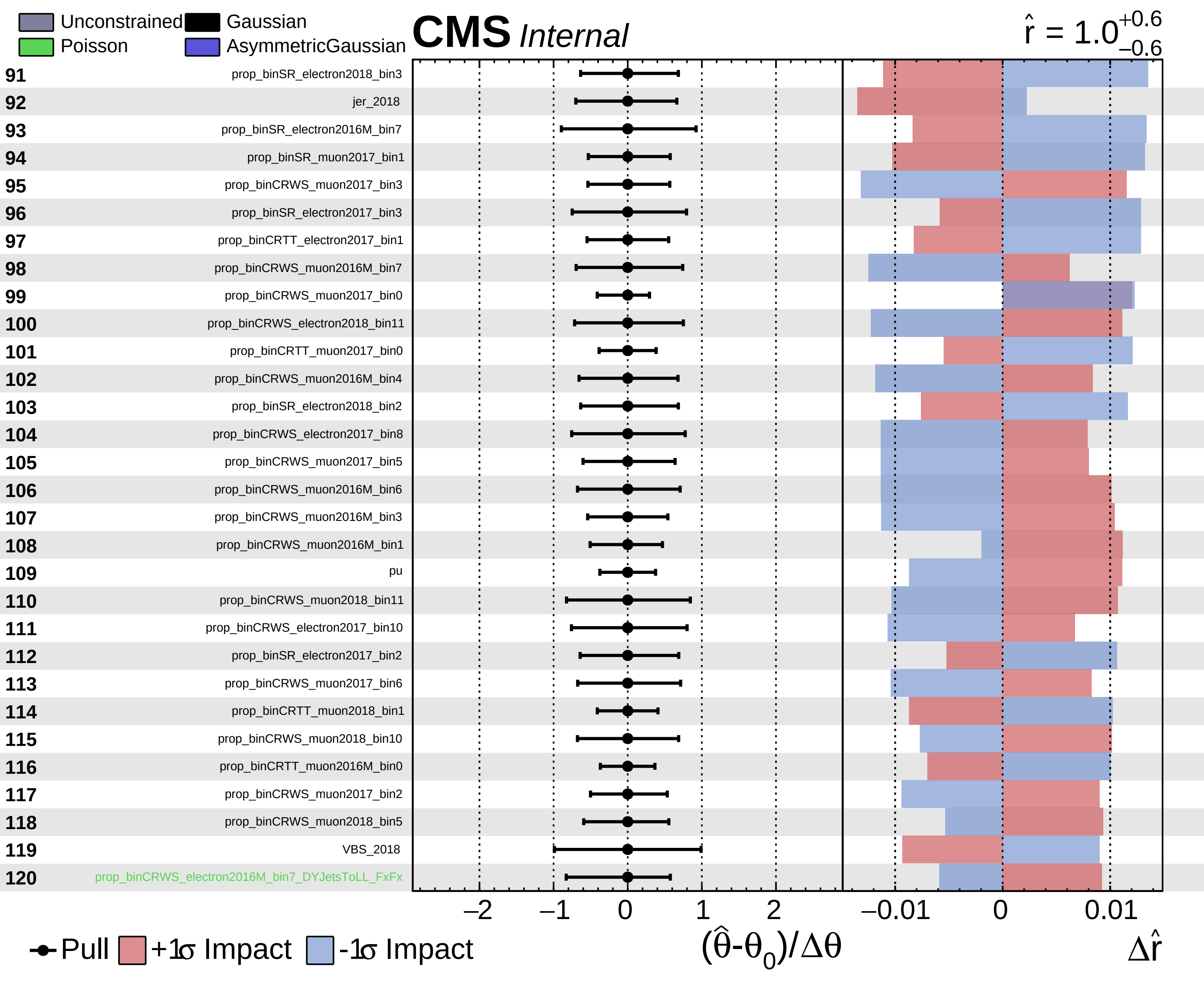

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
1 prop_binSR_muon2018_bin12
2 QCDScale_sig
3 prop_binSR_electron2018_bin12
4 prop_binSR_muon2017_bin12
5 prop_binSR_muon2018_bin11
6 prop_binSR_electron2017_bin12
7 prop_binSR_muon2018_bin10
8 pdf_Tot
9 prop_binSR_electron2018_bin11
10 prop_binSR_muon2016M_bin12
11 prop_binSR_muon2018_bin9
12 prop_binSR_muon2017_bin11
13 prop_binSR_electron2016M_bin12
14 prop_binSR_muon2018_bin6
15 prop_binSR_electron2018_bin10
16 prop_binSR_muon2017_bin10
17 prop_binSR_muon2016M_bin11
18 QCDScale_DY
19 prop_binSR_muon2018_bin8
20 prop_binSR_muon2018_bin7
21 prop_binSR_muon2018_bin5
22 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
23 prop_binSR_electron2017_bin11
24 prop_binSR_muon2017_bin6
25 prop_binSR_muon2016M_bin10
26 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
27 prop_binSR_muon2017_bin9
28 prop_binSR_electron2018_bin6
29 prop_binSR_electron2018_bin9
30 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



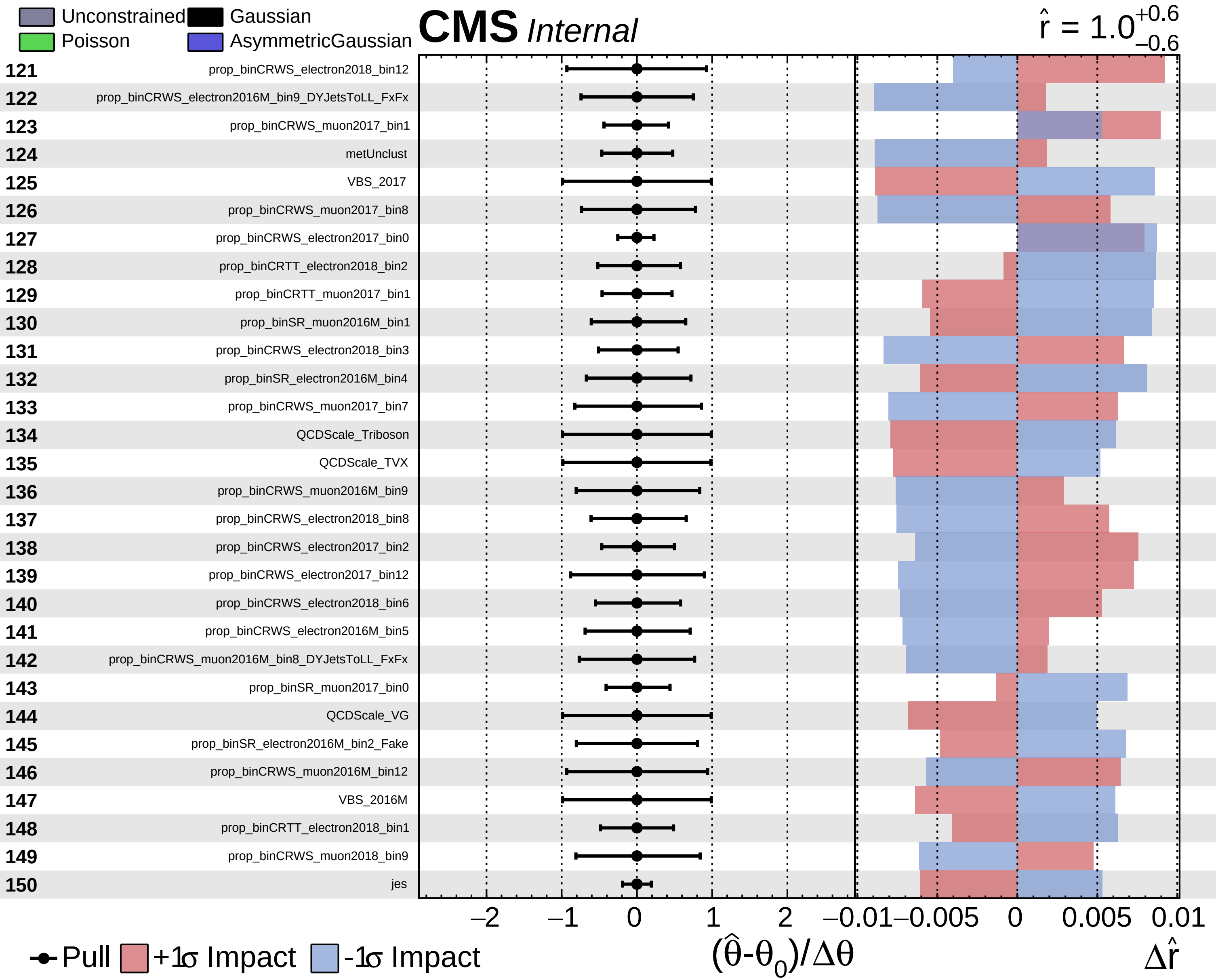
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
31 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
32 prop_binSR_muon2017_bin8
33 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
34 prop_binSR_electron2016M_bin11
35 prop_binSR_electron2017_bin10
36 prop_binSR_muon2016M_bin6
37 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
38 prop_binSR_muon2016M_bin9
39 prop_binSR_electron2018_bin8
40 lumi_2018
41 prop_binCRTT_muon2018_bin0
42 prop_binSR_muon2017_bin5
43 prop_binSR_electron2017_bin6
44 PF
45 prop_binCRWS_electron2018_bin2
46 prop_binSR_muon2018_bin4
47 prop_binCRWS_muon2016M_bin0
48 prop_binCRWS_muon2016M_bin2
49 prop_binSR_muon2017_bin4
50 prop_binSR_muon2017_bin7
51 prop_binSR_muon2018_bin3
52 prop_binCRWS_muon2018_bin3
53 prop_binSR_muon2018_bin2
54 prop_binSR_electron2018_bin7
55 prop_binSR_electron2018_bin5
56 prop_binSR_muon2016M_bin5
57 prop_binSR_electron2016M_bin10
58 prop_binSR_muon2016M_bin8
59 lumi_2017
60 prop_binSR_muon2016M_bin7
–2 –1 0 1 2
–0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

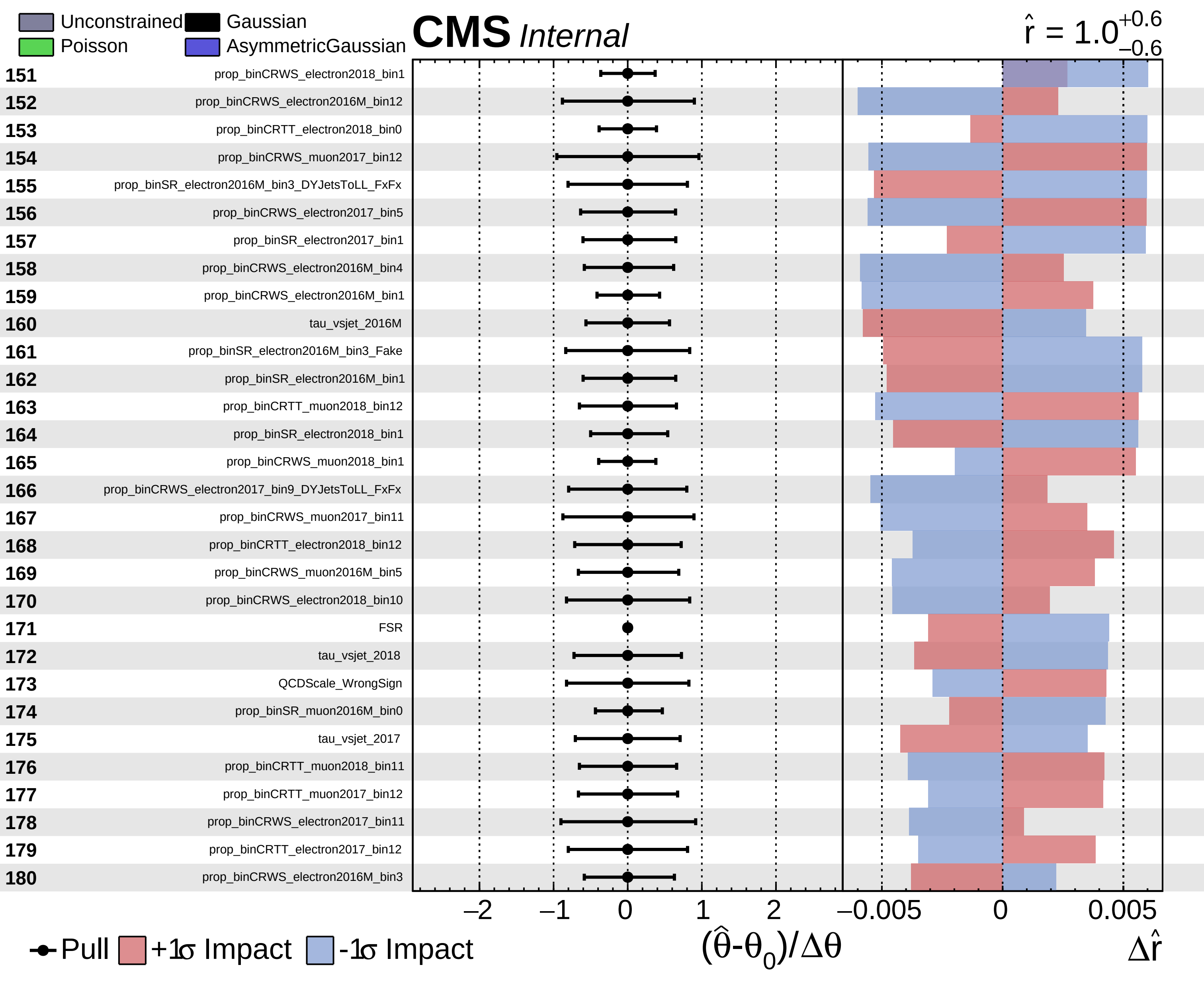


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
61 prop_binSR_electron2017_bin9
62 prop_binSR_electron2018_bin4
63 prop_binSR_electron2017_bin5
64 prop_binSR_electron2016M_bin9
65 prop_binCRWS_electron2018_bin5
66 prop_binSR_muon2017_bin3
67 prop_binSR_electron2016M_bin6
68 prop_binSR_electron2017_bin8
69 prop_binCRWS_electron2018_bin0
70 prop_binSR_muon2016M_bin4
71 prop_binSR_muon2016M_bin3
72 prop_binCRWS_electron2018_bin7
73 prop_binCRWS_muon2018_bin8
74 lumi_2016M
75 prop_binSR_muon2017_bin2
76 QCDScale_WpWpJJ_QCD
77 prop_binCRWS_muon2018_bin6
78 prop_binCRWS_electron2016M_bin0
79 prop_binSR_electron2017_bin7
80 btag
81 prop_binCRWS_muon2018_bin0
82 prop_binSR_electron2017_bin4
83 prop_binCRWS_muon2018_bin12
84 prop_binSR_electron2016M_bin5
85 prop_binCRWS_electron2017_bin4
86 prop_binSR_muon2016M_bin2
87 prop_binSR_electron2016M_bin8
88 prop_binSR_muon2018_bin1
89 prop_binCRWS_muon2017_bin4
90 mistag
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

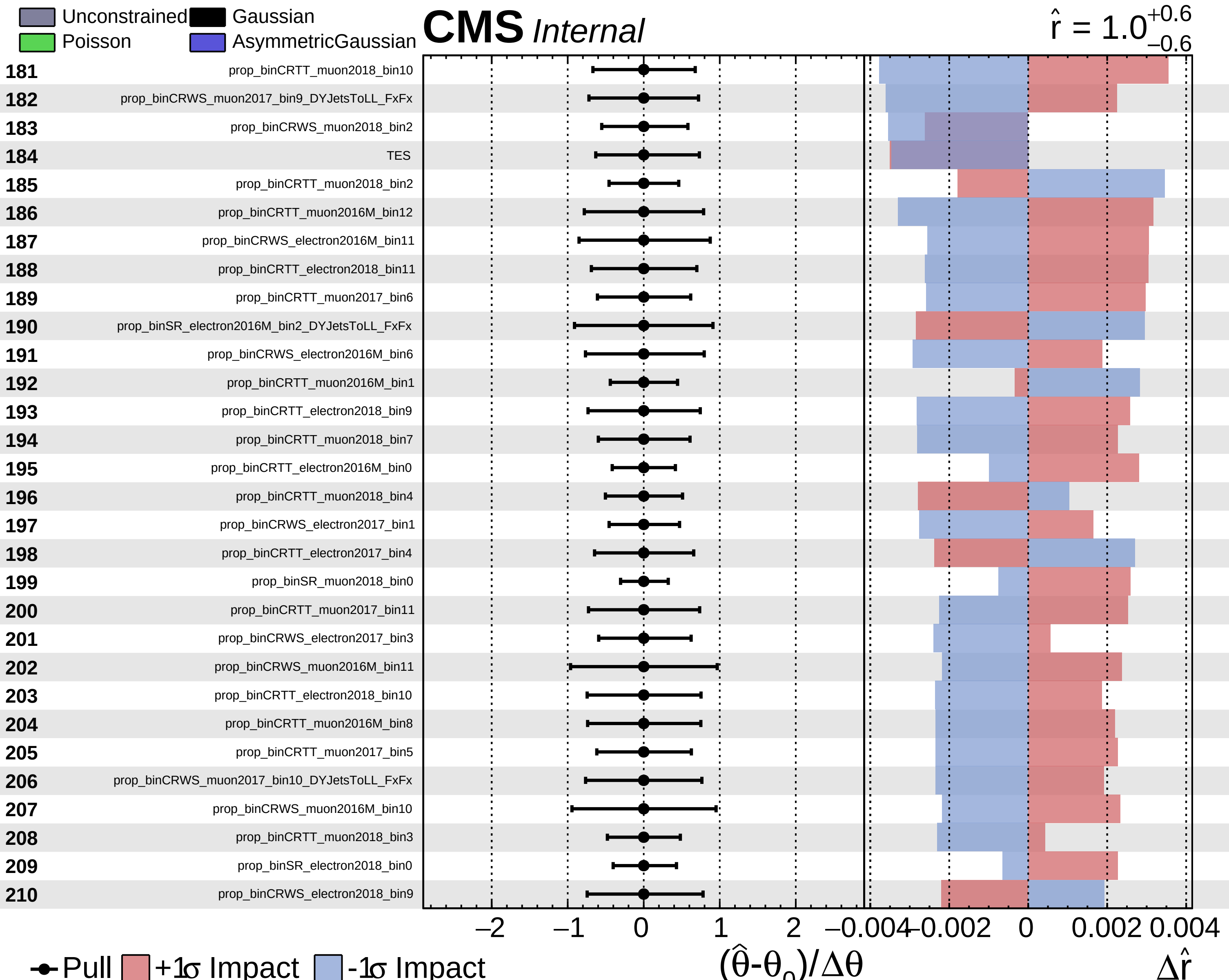


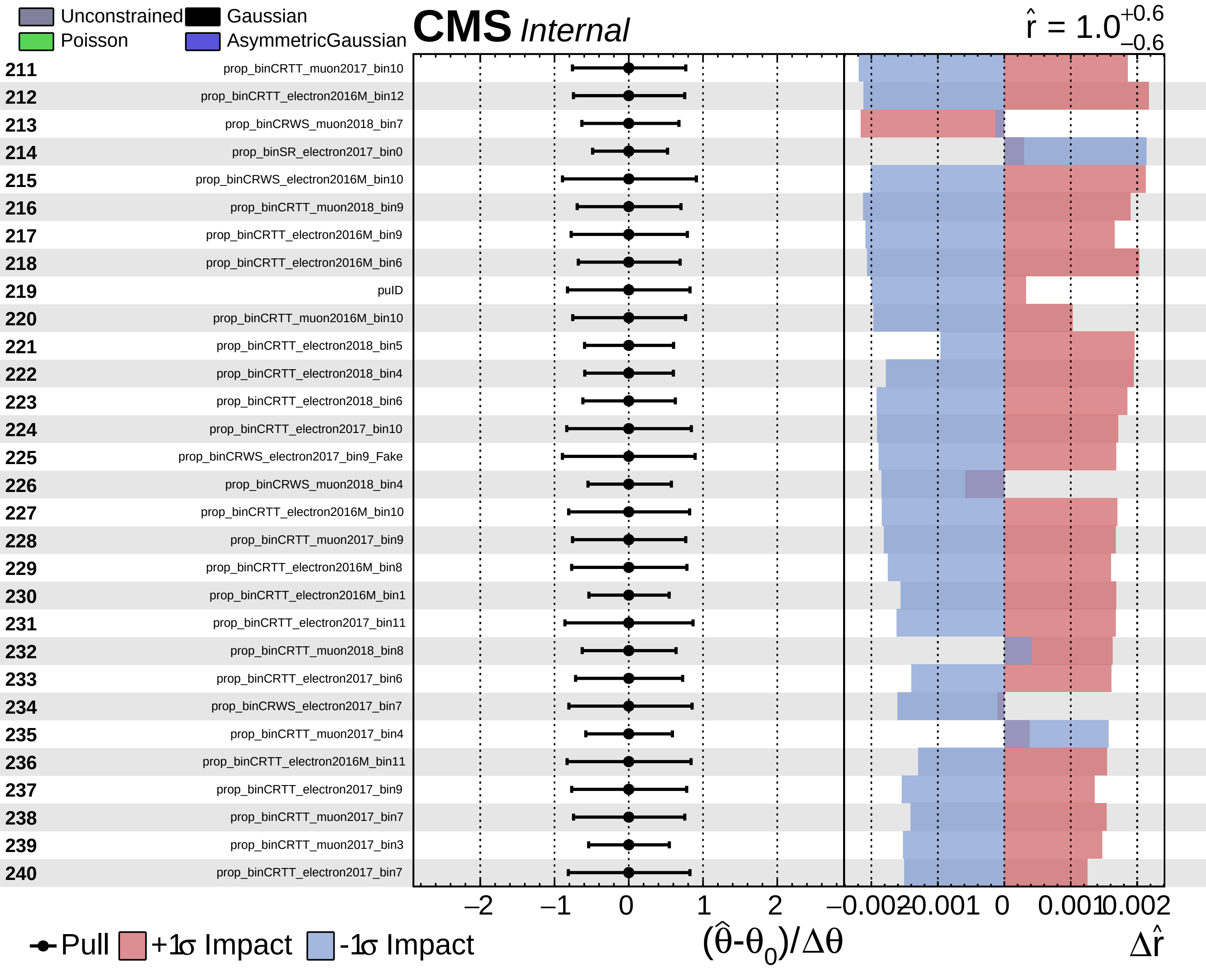
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.6
91 prop_binSR_electron2018_bin3
92 jer_2018
93 prop_binSR_electron2016M_bin7
94 prop_binSR_muon2017_bin1
95 prop_binCRWS_muon2017_bin3
96 prop_binSR_electron2017_bin3
97 prop_binCRTT_electron2017_bin1
98 prop_binCRWS_muon2016M_bin7
99 prop_binCRWS_muon2017_bin0
100 prop_binCRWS_electron2018_bin11
101 prop_binCRTT_muon2017_bin0
102 prop_binCRWS_muon2016M_bin4
103 prop_binSR_electron2018_bin2
104 prop_binCRWS_electron2017_bin8
105 prop_binCRWS_muon2017_bin5
106 prop_binCRWS_muon2016M_bin6
107 prop_binCRWS_muon2016M_bin3
108 prop_binCRWS_muon2016M_bin1
109 pu
110 prop_binCRWS_muon2018_bin11
111 prop_binCRWS_electron2017_bin10
112 prop_binSR_electron2017_bin2
113 prop_binCRWS_muon2017_bin6
114 prop_binCRTT_muon2018_bin1
115 prop_binCRWS_muon2018_bin10
116 prop_binCRTT_muon2016M_bin0
117 prop_binCRWS_muon2017_bin2
118 prop_binCRWS_muon2018_bin5
119 VBS_2018
120 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
121 prop_binCRWS_electron2018_bin12
122 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
123 prop_binCRWS_muon2017_bin1
124 metUnclust
125 VBS_2017
126 prop_binCRWS_muon2017_bin8
127 prop_binCRWS_electron2017_bin0
128 prop_binCRTT_electron2018_bin2
129 prop_binCRTT_muon2017_bin1
130 prop_binSR_muon2016M_bin1
131 prop_binCRWS_electron2018_bin3
132 prop_binSR_electron2016M_bin4
133 prop_binCRWS_muon2017_bin7
134 QCDScale_Triboson
135 QCDScale_TVX
136 prop_binCRWS_muon2016M_bin9
137 prop_binCRWS_electron2018_bin8
138 prop_binCRWS_electron2017_bin2
139 prop_binCRWS_electron2017_bin12
140 prop_binCRWS_electron2018_bin6
141 prop_binCRWS_electron2016M_bin5
142 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
143 prop_binSR_muon2017_bin0
144 QCDScale_VG
145 prop_binSR_electron2016M_bin2_Fake
146 prop_binCRWS_muon2016M_bin12
147 VBS_2016M
148 prop_binCRTT_electron2018_bin1
149 prop_binCRWS_muon2018_bin9
150 jes
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

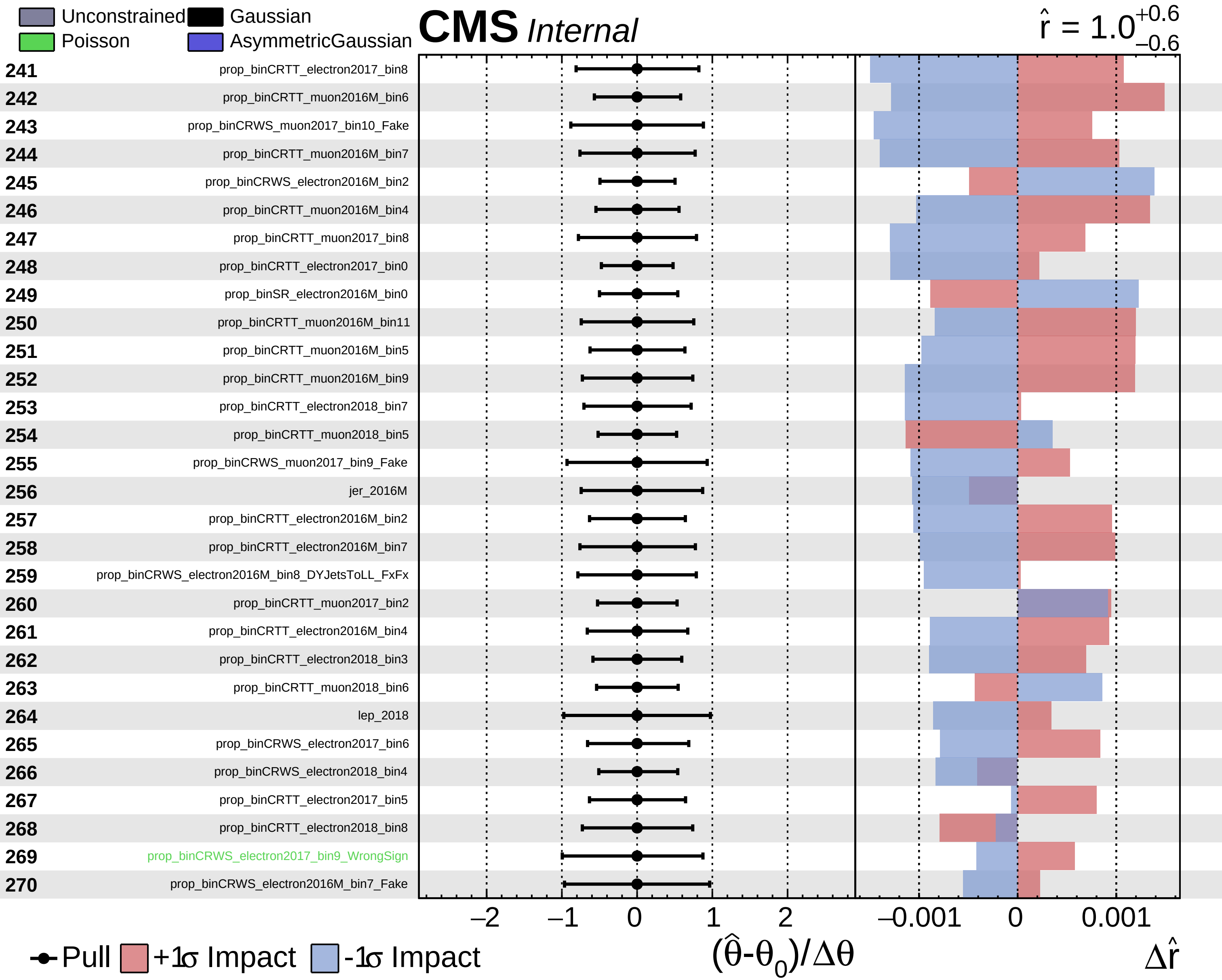


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.6
151 prop_binCRWS_electron2018_bin1
152 prop_binCRWS_electron2016M_bin12
153 prop_binCRTT_electron2018_bin0
154 prop_binCRWS_muon2017_bin12
155 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
156 prop_binCRWS_electron2017_bin5
157 prop_binSR_electron2017_bin1
158 prop_binCRWS_electron2016M_bin4
159 prop_binCRWS_electron2016M_bin1
160 tau_vsjet_2016M
161 prop_binSR_electron2016M_bin3_Fake
162 prop_binSR_electron2016M_bin1
163 prop_binCRTT_muon2018_bin12
164 prop_binSR_electron2018_bin1
165 prop_binCRWS_muon2018_bin1
166 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
167 prop_binCRWS_muon2017_bin11
168 prop_binCRTT_electron2018_bin12
169 prop_binCRWS_muon2016M_bin5
170 prop_binCRWS_electron2018_bin10
171 FSR
172 tau_vsjet_2018
173 QCDScale_WrongSign
174 prop_binSR_muon2016M_bin0
175 tau_vsjet_2017
176 prop_binCRTT_muon2018_bin11
177 prop_binCRTT_muon2017_bin12
178 prop_binCRWS_electron2017_bin11
179 prop_binCRTT_electron2017_bin12
180 prop_binCRWS_electron2016M_bin3
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

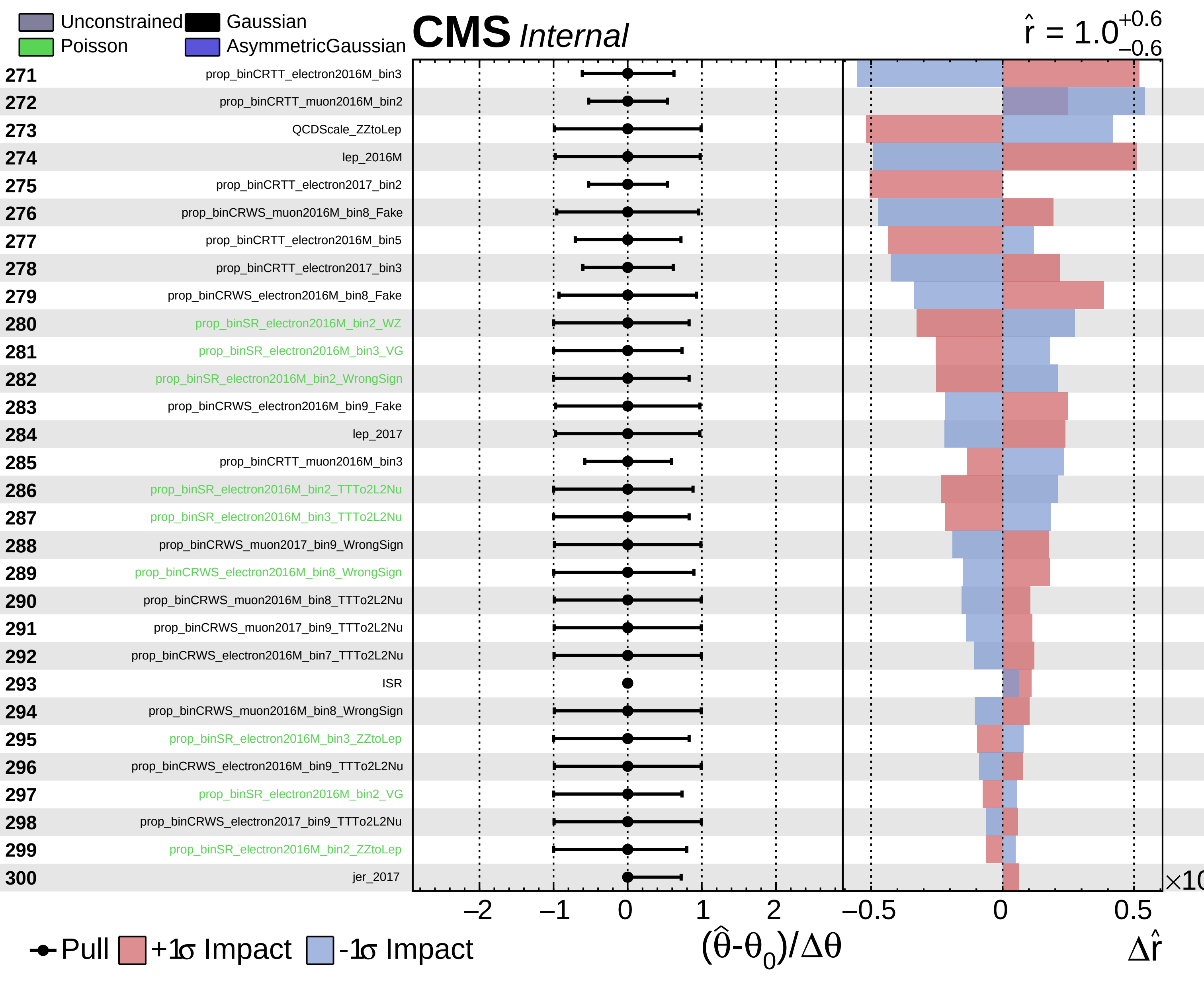

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
181 prop_binCRTT_muon2018_bin10
182 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
183 prop_binCRWS_muon2018_bin2
184 TES
185 prop_binCRTT_muon2018_bin2
186 prop_binCRTT_muon2016M_bin12
187 prop_binCRWS_electron2016M_bin11
188 prop_binCRTT_electron2018_bin11
189 prop_binCRTT_muon2017_bin6
190 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
191 prop_binCRWS_electron2016M_bin6
192 prop_binCRTT_muon2016M_bin1
193 prop_binCRTT_electron2018_bin9
194 prop_binCRTT_muon2018_bin7
195 prop_binCRTT_electron2016M_bin0
196 prop_binCRTT_muon2018_bin4
197 prop_binCRWS_electron2017_bin1
198 prop_binCRTT_electron2017_bin4
199 prop_binSR_muon2018_bin0
200 prop_binCRTT_muon2017_bin11
201 prop_binCRWS_electron2017_bin3
202 prop_binCRWS_muon2016M_bin11
203 prop_binCRTT_electron2018_bin10
204 prop_binCRTT_muon2016M_bin8
205 prop_binCRTT_muon2017_bin5
206 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
207 prop_binCRWS_muon2016M_bin10
208 prop_binCRTT_muon2018_bin3
209 prop_binSR_electron2018_bin0
210 prop_binCRWS_electron2018_bin9
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



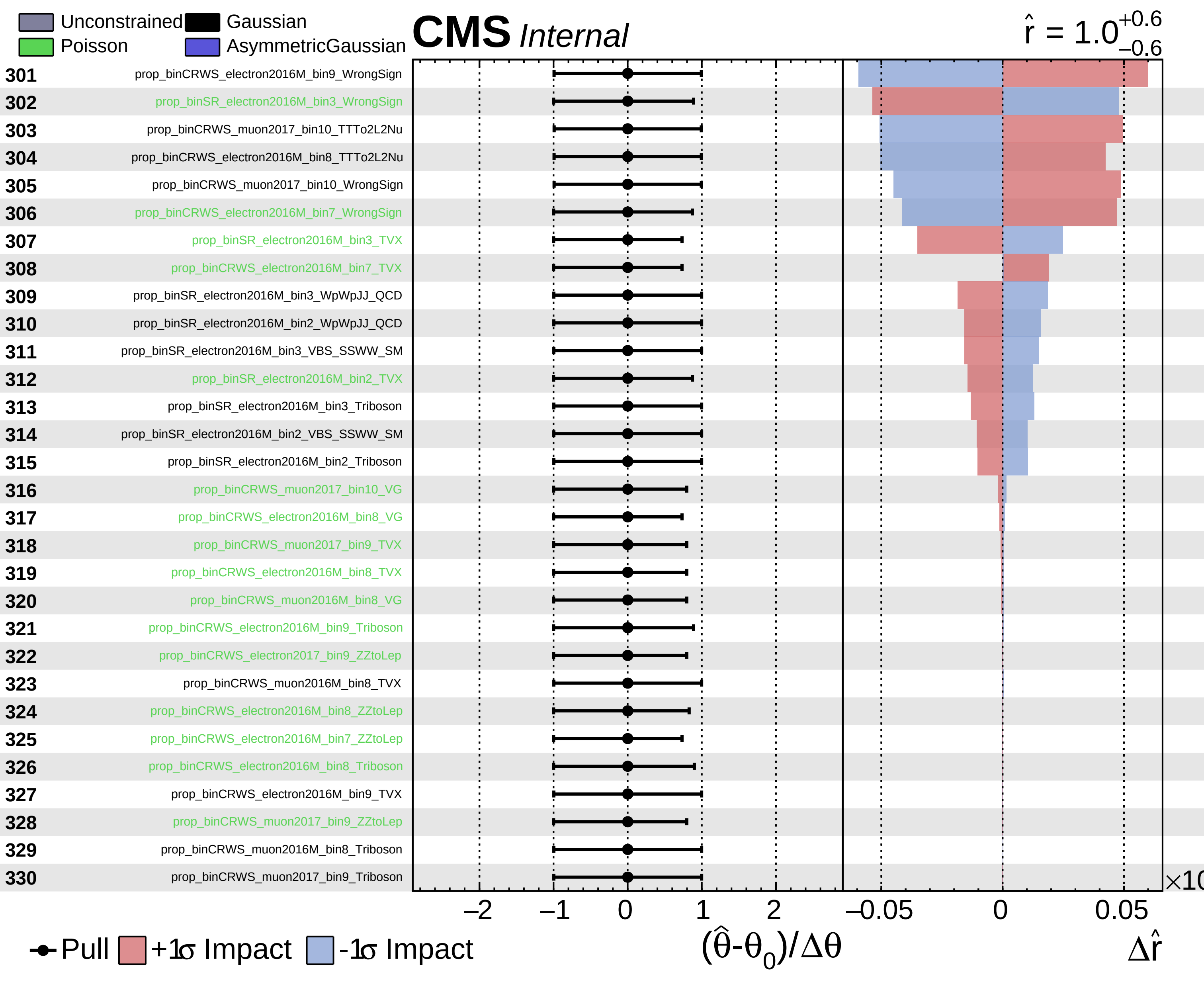
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
211 prop_binCRTT_muon2017_bin10
212 prop_binCRTT_electron2016M_bin12
213 prop_binCRWS_muon2018_bin7
214 prop_binSR_electron2017_bin0
215 prop_binCRWS_electron2016M_bin10
216 prop_binCRTT_muon2018_bin9
217 prop_binCRTT_electron2016M_bin9
218 prop_binCRTT_electron2016M_bin6
219 puID
220 prop_binCRTT_muon2016M_bin10
221 prop_binCRTT_electron2018_bin5
222 prop_binCRTT_electron2018_bin4
223 prop_binCRTT_electron2018_bin6
224 prop_binCRTT_electron2017_bin10
225 prop_binCRWS_electron2017_bin9_Fake
226 prop_binCRWS_muon2018_bin4
227 prop_binCRTT_electron2016M_bin10
228 prop_binCRTT_muon2017_bin9
229 prop_binCRTT_electron2016M_bin8
230 prop_binCRTT_electron2016M_bin1
231 prop_binCRTT_electron2017_bin11
232 prop_binCRTT_muon2018_bin8
233 prop_binCRTT_electron2017_bin6
234 prop_binCRWS_electron2017_bin7
235 prop_binCRTT_muon2017_bin4
236 prop_binCRTT_electron2016M_bin11
237 prop_binCRTT_electron2017_bin9
238 prop_binCRTT_muon2017_bin7
239 prop_binCRTT_muon2017_bin3
240 prop_binCRTT_electron2017_bin7
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

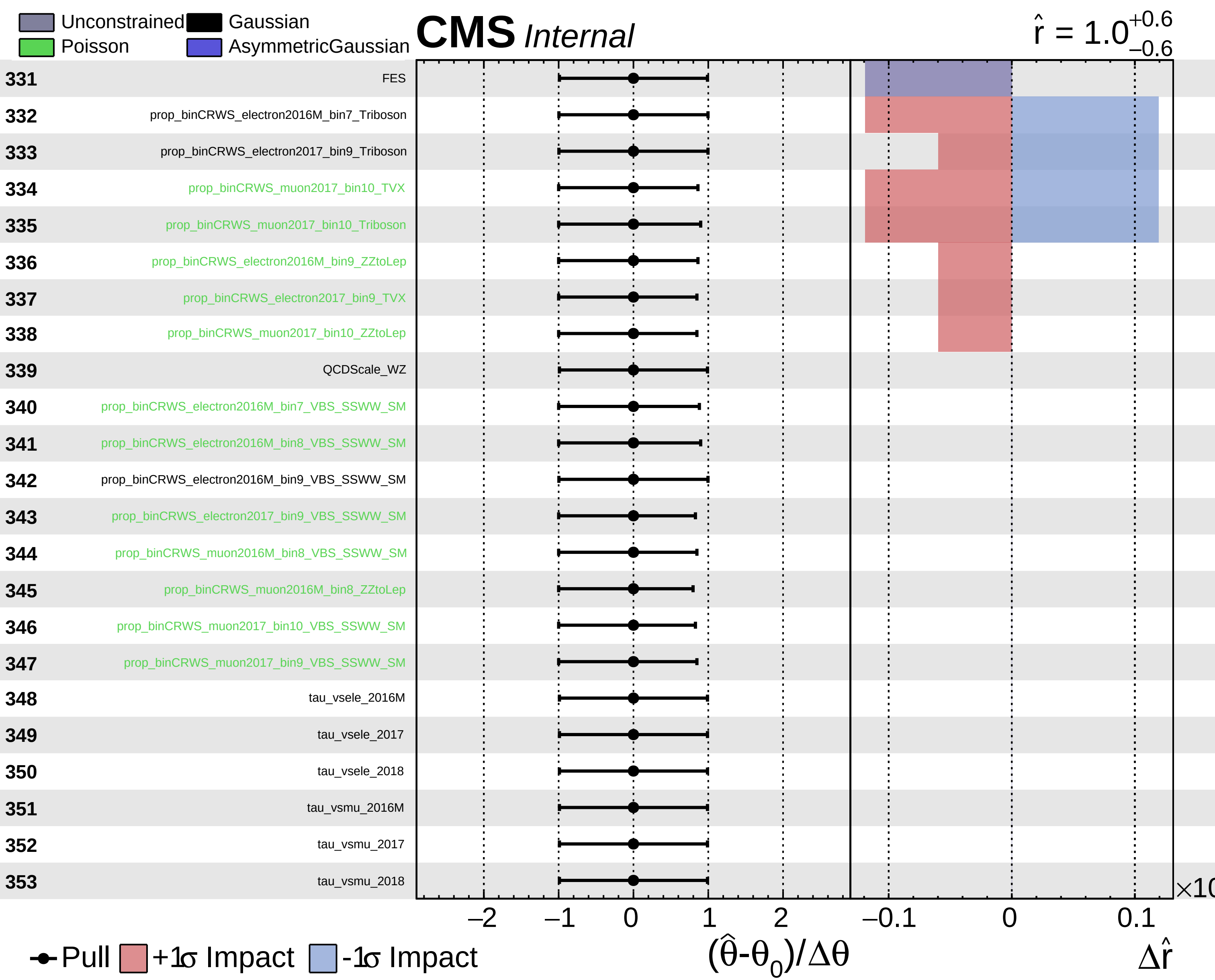


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
241 prop_binCRTT_electron2017_bin8
242 prop_binCRTT_muon2016M_bin6
243 prop_binCRWS_muon2017_bin10_Fake
244 prop_binCRTT_muon2016M_bin7
245 prop_binCRWS_electron2016M_bin2
246 prop_binCRTT_muon2016M_bin4
247 prop_binCRTT_muon2017_bin8
248 prop_binCRTT_electron2017_bin0
249 prop_binSR_electron2016M_bin0
250 prop_binCRTT_muon2016M_bin11
251 prop_binCRTT_muon2016M_bin5
252 prop_binCRTT_muon2016M_bin9
253 prop_binCRTT_electron2018_bin7
254 prop_binCRTT_muon2018_bin5
255 prop_binCRWS_muon2017_bin9_Fake
256 jer_2016M
257 prop_binCRTT_electron2016M_bin2
258 prop_binCRTT_electron2016M_bin7
259 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
260 prop_binCRTT_muon2017_bin2
261 prop_binCRTT_electron2016M_bin4
262 prop_binCRTT_electron2018_bin3
263 prop_binCRTT_muon2018_bin6
264 lep_2018
265 prop_binCRWS_electron2017_bin6
266 prop_binCRWS_electron2018_bin4
267 prop_binCRTT_electron2017_bin5
268 prop_binCRTT_electron2018_bin8
269 prop_binCRWS_electron2017_bin9_WrongSign
270 prop_binCRWS_electron2016M_bin7_Fake
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
271 prop_binCRTT_electron2016M_bin3
272 prop_binCRTT_muon2016M_bin2
273 QCDScale_ZZtoLep
274 lep_2016M
275 prop_binCRTT_electron2017_bin2
276 prop_binCRWS_muon2016M_bin8_Fake
277 prop_binCRTT_electron2016M_bin5
278 prop_binCRTT_electron2017_bin3
279 prop_binCRWS_electron2016M_bin8_Fake
280 prop_binSR_electron2016M_bin2_WZ
281 prop_binSR_electron2016M_bin3_VG
282 prop_binSR_electron2016M_bin2_WrongSign
283 prop_binCRWS_electron2016M_bin9_Fake
284 lep_2017
285 prop_binCRTT_muon2016M_bin3
286 prop_binSR_electron2016M_bin2_TTTo2L2Nu
287 prop_binSR_electron2016M_bin3_TTTo2L2Nu
288 prop_binCRWS_muon2017_bin9_WrongSign
289 prop_binCRWS_electron2016M_bin8_WrongSign
290 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
291 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
292 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
293 ISR
294 prop_binCRWS_muon2016M_bin8_WrongSign
295 prop_binSR_electron2016M_bin3_ZZtoLep
296 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
297 prop_binSR_electron2016M_bin2_VG
298 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
299 prop_binSR_electron2016M_bin2_ZZtoLep
300 jer_2017
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
301 prop_binCRWS_electron2016M_bin9_WrongSign
302 prop_binSR_electron2016M_bin3_WrongSign
303 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
304 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
305 prop_binCRWS_muon2017_bin10_WrongSign
306 prop_binCRWS_electron2016M_bin7_WrongSign
307 prop_binSR_electron2016M_bin3_TVX
308 prop_binCRWS_electron2016M_bin7_TVX
309 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
310 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
311 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
312 prop_binSR_electron2016M_bin2_TVX
313 prop_binSR_electron2016M_bin3_Triboson
314 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
315 prop_binSR_electron2016M_bin2_Triboson
316 prop_binCRWS_muon2017_bin10_VG
317 prop_binCRWS_electron2016M_bin8_VG
318 prop_binCRWS_muon2017_bin9_TVX
319 prop_binCRWS_electron2016M_bin8_TVX
320 prop_binCRWS_muon2016M_bin8_VG
321 prop_binCRWS_electron2016M_bin9_Triboson
322 prop_binCRWS_electron2017_bin9_ZZtoLep
323 prop_binCRWS_muon2016M_bin8_TVX
324 prop_binCRWS_electron2016M_bin8_ZZtoLep
325 prop_binCRWS_electron2016M_bin7_ZZtoLep
326 prop_binCRWS_electron2016M_bin8_Triboson
327 prop_binCRWS_electron2016M_bin9_TVX
328 prop_binCRWS_muon2017_bin9_ZZtoLep
329 prop_binCRWS_muon2016M_bin8_Triboson
330 prop_binCRWS_muon2017_bin9_Triboson
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.6}$
331 FES
332 prop_binCRWS_electron2016M_bin7_Triboson
333 prop_binCRWS_electron2017_bin9_Triboson
334 prop_binCRWS_muon2017_bin10_TVX
335 prop_binCRWS_muon2017_bin10_Triboson
336 prop_binCRWS_electron2016M_bin9_ZZtoLep
337 prop_binCRWS_electron2017_bin9_TVX
338 prop_binCRWS_muon2017_bin10_ZZtoLep
339 QCDScale_WZ
340 prop_binCRWS_electron2016M_bin7_VBS_SSWW_SM
341 prop_binCRWS_electron2016M_bin8_VBS_SSWW_SM
342 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
343 prop_binCRWS_electron2017_bin9_VBS_SSWW_SM
344 prop_binCRWS_muon2016M_bin8_VBS_SSWW_SM
345 prop_binCRWS_muon2016M_bin8_ZZtoLep
346 prop_binCRWS_muon2017_bin10_VBS_SSWW_SM
347 prop_binCRWS_muon2017_bin9_VBS_SSWW_SM
348 tau_vsele_2016M
349 tau_vsele_2017
350 tau_vsele_2018
351 tau_vsmu_2016M
352 tau_vsmu_2017
353 tau_vsmu_2018
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$